Do Now · retrieval

Noughts & Crosses · Part 1 — Sequence before loops

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a zero-indexed C# array, which element does array[2] access?

Q2

Name the three parts of a C# for loop header used to visit every array element.

Q3

In a C# 2D array organised as rows and columns, what does GetLength(0) return?

Q4

If a board has 3 rows and 3 columns, how many values does it contain?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start with sequence — Write the five lines in the left panel into a C# console program. Run it. There are no loops ; each statement executes in order.

Core idea 2

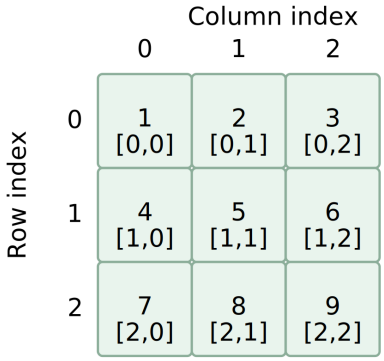
Make one simple change — Change the middle square 5 to X. Which exact line do you edit? This works — but changing the text every time a player moves is awkward. — CODE:
Console.WriteLine("4 | X | 6");

Core idea 3

Store the squares as data — Put the nine square values into a 2D array. The first index chooses the row; the second chooses the column. — CODE:

```
string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
};
```

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Display the data directly — Write the first row explicitly using three array accesses. Then do the same for rows 1 and 2. — CODE:

```
Console.WriteLine(board[0,0] + " | " +
board[0,1] + " | " +
board[0,2]);
```

Vocabulary · use the correct technical terms

sequence — Instructions that run in a specified order.

statement — One instruction in a program.

output — Information shown by the program.

2D array — Values stored using two indexes.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Make the board display before improving it

Follow the worked method

1. Create a working output — Run these instructions in order. The result is a 3×3 numbered board. — CODE:

```
Console.WriteLine("1 | 2 | 3");
Console.WriteLine("--+---+--");
Console.WriteLine("4 | 5 | 6");
Console.WriteLine("--+---+--");
Console.WriteLine("7 | 8 | 9");
```

2. Move the values into a 2D array — CODE:

```
string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
};
```

— Square 5 is board[1,1] , not board[5] , because our board uses zero-based row and column indexes.

3. Print the same values without a loop — CODE:

```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

— This code works. But how many times do the row-printing instructions repeat? What changes between lines? We'll tackle that in Lesson 014.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Write or explain

Without loops, write three C# statements that print 1 | 2 | 3, then 4 | 5 | 6, then 7 | 8 | 9.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Create a static board
In .NET Fiddle, write sequential Console.WriteLine() statements to display the nine squares. Do not use a loop.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Print the array without loops
Using three separate Console.WriteLine() statements and explicit board[row,col] accesses, display all three rows. Highlight the parts of the code that repeat.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

We must write a loop before we can display a 3×3 board in C#.

Use key vocabulary from memory

1. Explain sequence in your own words.

2. Explain statement in your own words.

3. Explain output in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.